

MAH-MCA-CET 2026

Official Course Syllabus & Exam Format Breakdown

The Online Common Entrance Test (CET) comprises 4 sections: Mathematics & Statistics, Logical / Abstract Reasoning, English Comprehension and Verbal Ability, and Computer Concepts. The exam consists of 100 questions of total 200 marks, with a composite duration of 90 minutes.

a) Mathematics & Statistics

Questions cover high school mathematics up to 12th standard:

- **Algebra:** Fundamental operations in Algebra, Expansion, factorization, Quadratic equations, indices, logarithms, arithmetic, geometric and harmonic progressions, binomial theorem, permutations and combinations.
- **Co-ordinate Geometry:** Rectangular Cartesian co-ordinates, equations of a line, mid point, intersections etc., equations of a circle, distance formulae, pair of straight lines, parabola, ellipse and hyperbola, simple geometric transformations such as translation, rotation, scaling.
- **Differential Equations:** Differential equations of first order and their solutions, linear differential equations with constant coefficients, homogeneous linear differential equations.
- **Trigonometry:** Simple identities, trigonometric equations, properties of triangles, solution of triangles, height and distance, inverse function.
- **Probability and Statistics:** Basic concepts of probability theory, Averages, Dependent and independent events, frequency distributions, and measures of dispersions, skewness and kurtosis, random variable and distribution functions, mathematical expectations, Binomial, Poisson, normal distributions, curve fitting, and principle of least squares, correlation and regression.
- **Arithmetic:** Ratios and proportions, problems on time-work, distance-speed, percentage, etc.
- **Basic Set Theory & Functions:** Set, relations and mappings.
- **Mensuration:** Areas, triangles and quadrilaterals, area and circumference of circles, volumes and surface areas of simple solids such as cubes, spheres, cylinders and cones.

b) Logical / Abstract Reasoning

This section includes questions to measure how quickly and logically you can think. It covers logical situations and questions based on the facts given in passages, evaluating the analytical and problem-solving capability of the candidate.

c) English Comprehension and Verbal Ability

Questions are designed to test the candidate's general understanding of the English language. Topics include Basic English grammar, vocabulary, comprehension, synonyms, antonyms, sentence correction, word & phrases, and jumbled paragraphs.

d) Computer Concepts

Covers foundational concepts in computing logic and organization:

- **Computer Basics:** Organization of a computer, Central Processing Unit (CPU), Structure of instructions in CPU, input / output devices, computer memory, memory organization, back-up devices.
- **Data Representation:** Representation of characters, integers, and fractions, binary and hexadecimal representations. Binary Arithmetic: Addition, subtraction, division, multiplication, signed arithmetic and two's complement arithmetic, floating point representation of numbers, normalized floating point representation, Boolean algebra, truth tables, Venn diagrams.
- **Computer Architecture:** Basics of Digital Logic, Block structure of computers, communication between processor and I/ O devices, interrupts.
- **Computer Language:** Fundamentals of Data & File Structures and high level language, Computer Programming in C, advanced concepts in programming.
- **Operating System:** Basics of operating systems, core processes, memory and resource management.

Exam Marking Scheme & Format

Section Topic	No. of Questions	Marks per Qs	Max Marks
Mathematics & Statistics	30	2	60
Logical / Abstract Reasoning	30	2	60
English Comprehension & Verbal	20	2	40
Computer Concepts	20	2	40
Total weightage	100	-	200

General Rules & Format:

- **Test Duration:** 90 Minutes composite time.
- **Negative Marking:** There is no negative marking system.
- **Medium of Exam:** English language only.
- **Mode:** Computer-based online examination.